

Day 5

1. Problem Definition

The project predicts whether an individual's annual income exceeds \$50K using the Adult Census Income dataset. The business framing prioritizes precision because a false-positive prediction is considered more costly than missing a true high-income case. The final Day 5 evaluation therefore focuses particularly on precision while still reporting recall, F1, ranking quality and probability quality.

2. Data Preparation and Validation

The final notebook loaded the Adult Census dataset from OpenML (adult, version 2) and recreated the deterministic stratified 70/10/20 train, development and untouched-test split using `random_state=42`. The resulting partitions were 34,188 training records, 4,885 development records and 9,769 held-out test records, with a positive-class rate of approximately 23.93% in each partition.

Feature engineering and preprocessing are contained in the saved Day 4 pipeline. This keeps transformations inside the model workflow rather than applying manual preprocessing to the test or inference data. The Day 5 notebook verified that the final hold-out set is separate from the training/development partitions.

3. Model Development and Tuning

Day 5 did not retrain or replace the tuned solution. Instead, it loaded the saved Day 4 artifact and validated the fitted pipeline directly. The loaded final estimator was a `CalibratedClassifierCV` object, and the saved classification threshold used for final predictions was 0.425.

During Day 4, candidate models were evaluated with stratified cross-validation and randomized hyperparameter search using precision as the primary optimization metric. The executed Day 5 artifact did not contain a shortlisted-model comparison table, so no additional comparison values are reported here rather than fabricating them.

4. Final Untouched-Test Results

Metric	Test value
Accuracy	0.859351
Precision	0.763676
Recall	0.597092
F1	0.670187
ROC-AUC	0.910646
PR-AUC	0.793375
Brier Score	0.099490

At the selected threshold of 0.425, the model achieved precision of 0.7637. This means that roughly 76.4% of records predicted as above \$50K were actually above \$50K. The ROC-AUC of 0.9106 and PR-AUC of 0.7934 indicate strong ranking performance, while the Brier score of 0.0995 provides a probability-quality measure.

5. Confusion Matrix and Error Analysis

TN	FP
FN	TP

Counts: TN = 6,999, FP = 432, FN = 942, TP = 1,396.

There were 432 false positives and 942 false negatives on the untouched test set. Under the project's business framing, the false positives are the costlier error type. The selected threshold therefore reflects the need to control positive-prediction quality rather than simply maximize accuracy.

The executed error analysis also surfaced high-confidence errors. High-confidence false positives included older private-sector workers with substantial capital gains, as well as several married respondents in managerial or professional occupations. High-confidence false negatives included younger workers and several respondents with HS-grad or Some-college education and little or no capital gain. These examples show that the model can encounter cases where individual attributes do not align cleanly with the income class.

6. Subgroup Analysis

The notebook generated subgroup performance for sex, race, education, workclass, marital status, occupation, age and hours-per-week, excluding groups with fewer than 30 observations. The output contained 130 subgroup rows. Performance was not uniform across groups; for example, workclass results in the executed analysis ranged from materially weaker performance for some groups to stronger performance for others. Age groups with very low positive prevalence also showed near-zero precision/recall, illustrating that subgroup metrics must be interpreted together with subgroup sample size and class prevalence.

7. Feature and Model Interpretation

The executed Day 5 interpretation cell did not find a supported `coefficient` or `feature_importances_` interface on the loaded `CalibratedClassifierCV` object and therefore correctly did not fabricate a feature importance ranking. This is a limitation of the current artifact structure, not evidence that the underlying model has no informative features. A future interpretation pass can inspect the calibrated estimator's fitted base estimators explicitly or use model-agnostic methods such as permutation importance or SHAP.

8. Production-Ready Inference

The inference workflow accepts raw input records and passes them directly through the saved pipeline. The pipeline performs the stored feature engineering and preprocessing automatically, produces a positive-class probability, and applies the saved 0.425 threshold to obtain the final prediction. The notebook successfully tested the inference function on 10 unseen examples from the untouched test set.

The ten example probabilities produced by the executed notebook were: 0.038434, 0.044800, 0.078206, 0.029020, 0.028990, 0.138463, 0.029261, 0.936708, 0.247950 and 0.788985. With the 0.425 threshold, the first seven and the ninth examples were predicted negative, while the eighth and tenth were predicted positive.

9. Limitations and Future Improvements

The Adult Census dataset is a benchmark dataset and may not represent a current production population. Historical demographic relationships can also introduce fairness concerns. Future work should include external validation, stakeholder-defined error costs, subgroup monitoring, drift detection, probability recalibration when necessary, and a more robust model-interpretation layer for the calibrated final estimator.

10. Reproducibility

The executed notebook recorded Python 3.13.15, NumPy 2.1.3, pandas 2.2.3, scikit-learn 1.6.1, joblib 1.5.3, matplotlib 3.10.0 and SciPy 1.16.3. `python-docx` was not installed in the executed environment, which is why this report is generated separately rather than by the Day 5 notebook.